

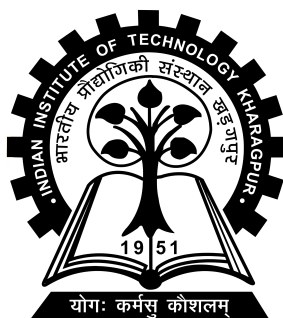
DPO for Medical VLMs: Efficient Chain-of-Thought Learning and Out-of-Domain Generalization

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TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Kolathuru Vishnu Vardhan**, a final-year Dual Degree student at the **Indian Institute of Technology, Kharagpur**, bearing Roll No. **21CE3AI13**, has successfully completed a Research Internship on the project titled “**DPO for Medical VLMs: Efficient Chain-of-Thought Learning and Out-of-Domain Generalization**” during the period **12th May 2025 to 20th August 2025**.

The internship was carried out under the joint supervision of faculty members from **Deakin University, Australia**, and **Shanghai Jiao Tong University, China**.

We hereby acknowledge his contributions to the project and extend our best wishes for his future endeavors.

Best regards,

Dr. Chaoxiang He

A handwritten signature in black ink, appearing to read '贺超翔' (He Chaoxiang).

Research Assistant Professor, Shanghai Jiao Tong University
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ABSTRACT

Large Vision-Language Models (VLMs) have shown strong performance in medical visual question answering (MedVQA) [10][11], but they often struggle to generalize reasoning across unseen modalities [2]. In this work, we replicate the MedVLM-R1 pipeline [1] with the goal of improving medical reasoning through chain-of-thought (CoT) explanations. Our approach builds on the principle that “Supervised Fine-Tuning (SFT) memorizes, whereas Reinforcement Learning (RL) generalizes” [4]. We first apply SFT on 1,000 in-domain MRI samples from the OmniMedVQA dataset [11] using Qwen2-VL-2B-Instruct [7] to establish generic response patterns. We then employ Direct Preference Optimization (DPO) [3], where high-quality CoT responses from MedVLM-R1 [1] serve as preferred outputs and SFT responses as rejected ones. This setup enables effective reasoning alignment without the computational burden or reward-model dependencies of GRPO [3][4].

Evaluation on in-domain MRI and out-of-domain CT and X-ray shows that our DPO-trained model achieves **99.3%** and **91.3%** accuracy, respectively, matching MedVLM-R1 benchmarks [1] while reducing training time by **~26x** (9 minutes vs. 4 hours on 2×A100 GPUs). The DPO+LoRA pipeline [5][6] is memory-efficient, fast, and capable of producing clinically meaningful CoT explanations that enhance interpretability. Our findings demonstrate that lightweight preference-based RL strategies can significantly improve multimodal reasoning generalization [4], making VLMs more practical for real-world medical AI [9][12].

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1. Introduction

Vision-Language Models (VLMs) are increasingly applied to medical tasks such as Medical Visual Question Answering (MedVQA) [10][11], where they reason over both medical images and natural language queries. While these models perform well on in-domain datasets, they often fail to generalize reasoning to out-of-domain modalities like Computed Tomography (CT) or X-ray, limiting their clinical utility [2][9].

Recent work has shown that Supervised Fine-Tuning (SFT) enables models to learn task-specific response patterns but often leads to memorization rather than genuine reasoning [4]. In contrast, Reinforcement Learning (RL) encourages structured reasoning and improves generalization [3][4]. For example, the MedVLM-R1 study [1] used Group Relative Policy Optimization (GRPO) to enhance reasoning in medical VLMs, achieving strong accuracy. However, GRPO is computationally expensive, depends on engineered reward functions, and introduces complexities for multimodal training [3][4].

To address these limitations, we explore an alternative – Direct Preference Optimization (DPO) [3], which learns directly from preference pairs instead of relying on explicit reward models. In this setup, high-quality chain-of-thought (CoT) explanations serve as “chosen” responses, while generic outputs are treated as “rejected.” This preference-based approach provides a simpler and more efficient way to guide medical reasoning [1][3].

Our contributions are as follows:

- **Efficient fine-tuning pipeline:** We perform SFT on 1,000 in-domain MRI samples from the OmniMedVQA dataset [11], teaching the model general response formats before applying DPO.
- **Preference-based reinforcement learning:** Using high-quality CoT explanations generated from the MedVLM-R1 checkpoint [1] as “chosen” responses and generic SFT outputs as “rejected” responses, we train a VLM (Qwen2-VL-2B-Instruct) via DPO [7].
- **Improved generalization:** Experimental results show that our approach achieves 99.33% in-domain accuracy (MRI) and 91.33% out-of-domain accuracy (CT and X-ray), comparable to MedVLM-R1 [1], while requiring only ~9 minutes of training on 2×A100 GPUs.
- **Practical applicability:** The DPO + LoRA pipeline [5][6] is memory- and compute-efficient, reducing training time by ~26x compared to GRPO-based methods [1][3][4], and is deployable even in low-resource environments.

This study demonstrates that DPO, combined with LoRA optimization [5][6], can enhance reasoning in medical VLMs while reducing computational overhead, offering a practical pathway for clinical deployment [9][12].

2. Related Work

2.1. Medical VLMs and Their Limitations

The emergence of Vision-Language Models (VLMs) has opened new possibilities in healthcare, especially in Medical Visual Question Answering (MedVQA) [10][11]. Domain-specific adaptations such as LLaVA-Med [12], PMC-VQA [10], and HuatuoGPT-Vision [9] have shown strong performance in radiology reasoning, typically using Supervised Fine-Tuning (SFT) on curated medical datasets.

While effective, SFT faces key drawbacks including high data requirements where models often need hundreds of thousands to millions of samples [2], which are expensive and limited by privacy concerns, limited reasoning since final-answer supervision fails to capture intermediate reasoning and reduces interpretability for clinicians [2][11], and poor generalization, as SFT models overfit to training distributions and degrade on out-of-domain modalities such as CT or X-ray when trained on MRI [2][9][11]. These limitations motivate strategies that improve both accuracy and reasoning generalization.

2.2. Reinforcement Learning for Enhanced Reasoning

Reinforcement Learning (RL) has emerged as a promising solution by rewarding both correctness and reasoning quality [3][4]. Classical methods like Proximal Policy Optimization (PPO) demonstrate that while “SFT memorizes, RL generalizes” [4]. Building on PPO, Group Relative Policy Optimization (GRPO) was adopted in MedVLM-R1 [1], achieving stronger reasoning accuracy and consistency.

However, GRPO introduces challenges: reliance on reward engineering, high computational cost, and difficulty in extending to multimodal tasks [3][4].

2.3. Preference Optimization as an Alternative

Direct Preference Optimization (DPO) offers a simpler alternative by learning directly from preference pairs [3], favoring high-quality chain-of-thought responses over weaker ones, without explicit reward models. Combined with LoRA [5][6], DPO provides a stable, compute-efficient alignment method that is easier to adopt in resource-constrained environments.

Our work builds on these insights by contrasting GRPO-based MedVLM-R1 [1] with a DPO-based pipeline on Qwen2-VL-2B-Instruct [7], demonstrating that preference optimization achieves comparable reasoning performance at a fraction of the training cost.

3. Methodology

The objective of this work is to enhance the generalization and reasoning capabilities of Vision-Language Models (VLMs) in medical applications by leveraging reinforcement learning [2][3][4]. Our methodology follows a two-stage pipeline: (i) supervised fine-tuning (SFT) on domain-specific data, and (ii) reinforcement learning using Direct Preference Optimization (DPO) to align the model with reliable chain-of-thought (CoT) reasoning [1][3].

3.1. Model Selection

We adopted Qwen2-VL-2B-Instruct [7], a lightweight yet instruction-tuned VLM capable of processing multimodal inputs. Its manageable size makes it suitable for fine-tuning within limited compute budgets while retaining strong reasoning capabilities.

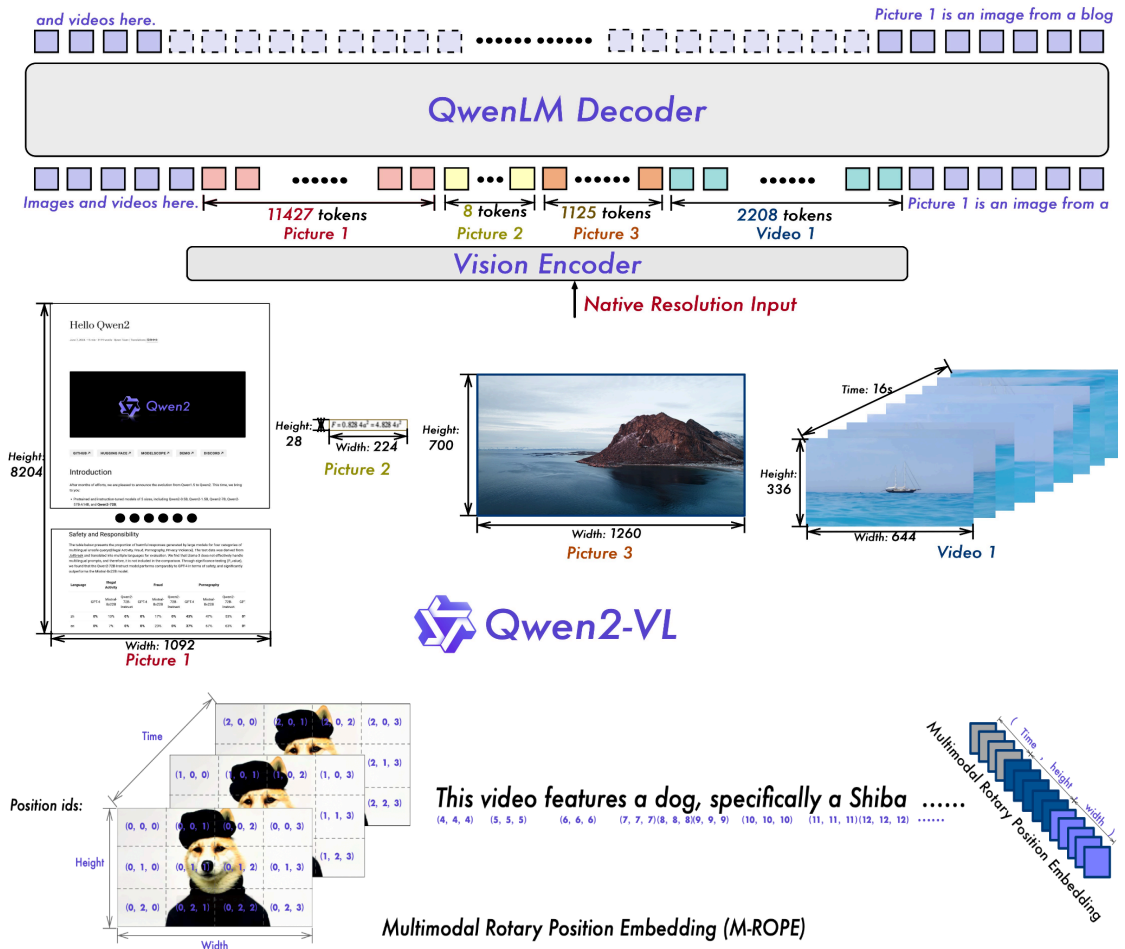


Figure 1. Overview of the adopted Qwen2-VL-2B-Instruct architecture [7]. The model integrates Naive Dynamic Resolution to dynamically tokenize images of arbitrary resolutions, enabling flexible processing of diverse medical modalities such as MRI, CT, and X-ray. Multimodal Rotary Position Embedding (M-ROPE) captures positional information across text (1D), images (2D), and videos (3D), enhancing cross-modal alignment.

3.2. Dataset

Experiments were conducted using the OmniMedVQA dataset [11], a benchmark for medical visual question answering that spans multiple imaging modalities (MRI, CT, and X-ray) paired with structured natural language questions and multiple-choice answers. Each instance includes an image, a question, four candidate answers, and one ground-truth label, enabling consistent evaluation of predictive accuracy and reasoning.

We selected OmniMedVQA because it covers diverse body regions with clear, well-annotated QA pairs [11], making it especially suitable for studying both in-domain performance (MRI) and out-of-domain generalization (CT and X-ray). For training, 1,100 MRI-based samples were used, split into 1,000 training and 100 validation examples, while CT and X-ray subsets were reserved exclusively for testing cross-modality generalization.

3.3. Prompt Format

To standardize reasoning outputs and facilitate evaluation [1][10], all models were trained and tested using the following response format:

1. *Generate step-by-step reasoning enclosed within `<think>...</think>` tags.*
2. *Provide the final single-letter answer (A, B, C, D, ...) inside `<answer>...</answer>` tags.*
3. *No additional text outside these tags.*

3.4. Stage 1: Supervised Fine-Tuning (SFT)

The first stage involved adapting the base model to the MRI domain. SFT was performed on 1,000 MRI training samples using LLaMA-Factory [5] with LoRA parameter-efficient fine-tuning [6]. Both the vision tower and multimodal projector were frozen to constrain training to language layers. Training was run on $2 \times$ NVIDIA A100 GPUs (40 GB each) for 5 epochs.

Key hyperparameters included: maximum sequence length = 2048, learning rate = 2×10^{-5} , batch size = 4, optimizer = AdamW, cosine learning rate scheduler with 5% warmup, and bfloat16 precision.

3.5. Stage 2: Reinforcement Learning with DPO

While SFT improved in-domain performance, it remained limited in generalization to unseen modalities [4]. To address this, we applied Direct Preference Optimization (DPO) [3]. Unlike GRPO (used in MedVLM-R1 [1]), which requires a reward model and extensive computation, DPO directly learns from preference pairs, making it more efficient and stable [3][4].

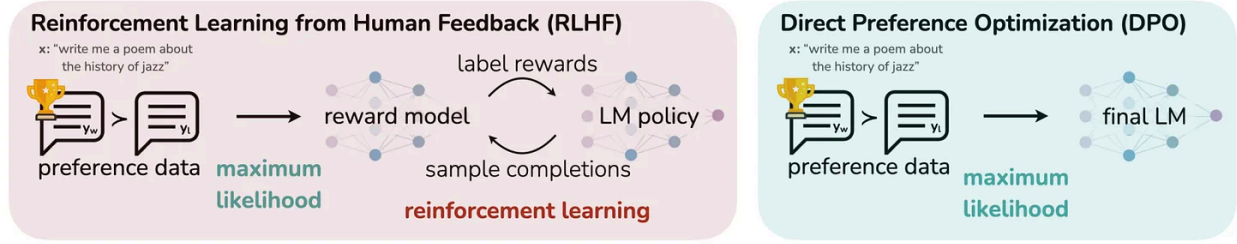


Figure 2. RLHF versus DPO training pipelines. RLHF first learns a reward model from preference data and then updates the LM policy via reinforcement learning. In contrast, DPO skips the reward model and online RL, optimizing the policy directly from preference pairs with a likelihood-style objective.

3.5.1. DPO Training Objective

The core of DPO is its training objective, which reframes preference learning as a binary classification task [3]. Instead of explicitly modeling a reward function, DPO optimizes the relative log-probabilities of preferred versus rejected responses in a single-stage policy optimization process. This improves stability and reduces computational overhead compared to RLHF.

Implicitly estimate reward using the policy

$$r(x, y) = \beta \log \frac{\pi_{\theta}(y | x)}{\pi_{\theta_{\text{ref}}}(y | x)}$$

Sigmoid function

DPO training loss

$$\mathcal{L}(\theta, \theta_{\text{ref}}) = -\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_c | x)}{\pi_{\theta_{\text{ref}}}(y_c | x)} - \beta \log \frac{\pi_{\theta}(y_r | x)}{\pi_{\theta_{\text{ref}}}(y_r | x)} \right)$$

LLM (policy) parameters Reference policy parameters Chosen reward Rejected reward KL divergence coefficient

Figure 3. The Direct Preference Optimization (DPO) training objective [3]. The loss function $\mathcal{L}(\theta, \theta_{\text{ref}})$ maximizes the log-likelihood of correctly classifying preferred responses by using an implicit reward function $r(x, y)$, which is parameterized by the policy π_{θ} and a frozen reference policy $\pi_{\theta_{\text{ref}}}$.

Key Components

- **Implicit Reward (r):** This is a calculation that measures how much the fine-tuned model (π_{θ}) prefers a response compared to the SFT-tuned baseline model ($\pi_{\theta_{\text{ref}}}$). A higher value means the fine-tuned model is much more likely to generate that response.

- **Chosen vs. Rejected Rewards:** The DPO loss directly compares the implicit reward for the chosen response (y_c) to the implicit reward for the rejected one (y_r). The model is trained to make the chosen reward larger than the rejected reward.
- **KL Divergence Coefficient (β):** This is a penalty factor. It ensures the fine-tuned model doesn't change too drastically from the SFT-tuned model. A higher β means the DPO model has to stay closer to its SFT behavior.
- **Sigmoid Function (σ):** This function is used to create a simple binary classification problem. The model's task is to "correctly classify" the chosen response as better than the rejected one. The loss is minimized when the chosen response has a much higher implicit reward.

3.5.2. Preference Pair Construction

Preference data was generated by importing the MedVLM-R1 checkpoint [1](Hugging Face: *JZPeterPan/MedVLM-R1*). For each MRI sample, two candidate responses were created:

Chosen: MedVLM-R1-generated CoT + correct answer (guaranteed correct).

Rejected: Generic SFT-style response without detailed reasoning (e.g., “*<think>Based on the image and options, the correct answer is C.</think><answer>C</answer>*”).

This ensured that the model learned to align with medically reliable CoT reasoning rather than surface-level predictions.

3.5.3. DPO Training Setup

DPO fine-tuning was performed on the SFT checkpoint using the same hardware setup. Key parameters included: learning rate = 2×10^{-5} , epochs = 5, batch size = 4, LoRA rank = 16 (α = 32, dropout = 0.1), preference β = 0.1, ftx = 0.2, and sigmoid loss. Here, β controls the weight of the preference loss, and ftx determines the fraction of preference pairs used per training step.

3.6. Evaluation Protocol

Model performance was evaluated on both in-domain (MRI) and out-of-domain (CT and X-ray) test sets using answer accuracy (percentage of correct multiple-choice predictions). Three configurations were compared:

1. Base model (zero-shot) without adaptation,
2. SFT model,
3. SFT + DPO model.

In addition, to ensure a fair comparison with prior work, we evaluated our DPO model on 300 MRI, 300 CT, and 300 X-ray test samples and compared it against the MedVLM-R1 benchmark results [1]. This analysis quantifies both generalization capability and computational efficiency.

4. Results & Discussion

4.1. Supervised Fine-Tuning (SFT)

The supervised fine-tuning stage was performed for 5 epochs on 1000 MRI training samples. The model converged rapidly, achieving a final training loss of **0.0655** after a runtime of 2 min 29.87 sec, while the evaluation loss stabilized at **0.0038** with an evaluation runtime of only 1.32 sec. These results indicate stable optimization and absence of overfitting. Similar observations about the strong memorization tendency of SFT have been reported in prior studies [4].

We also plot the training and evaluation loss curves in Figure 4, which show a smooth decline throughout training.

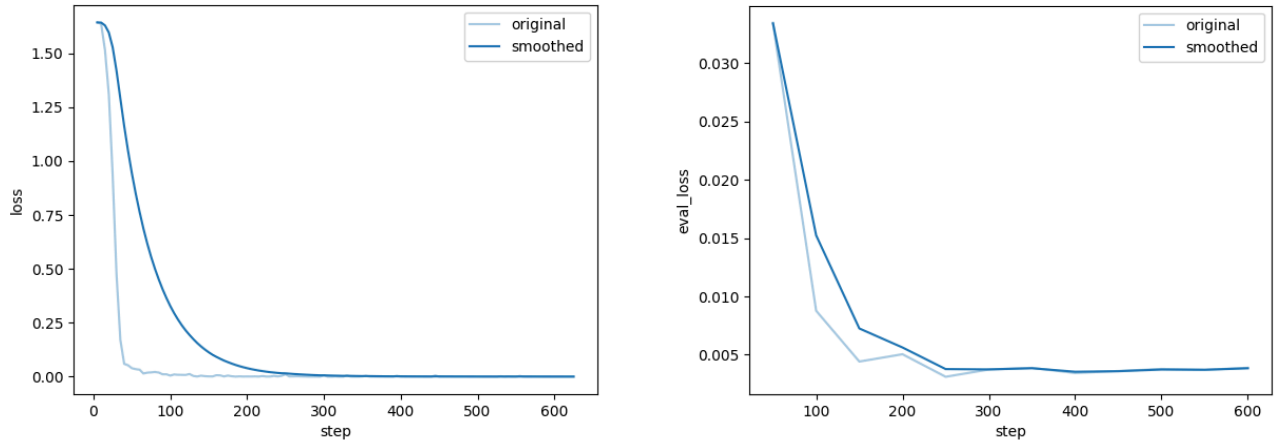


Figure 4. Training and evaluation loss curves for supervised fine-tuning. The plot on the left shows the training loss, which decreases rapidly and converges to a near-zero value, indicating effective learning. The plot on the right displays the evaluation loss, which also decreases and then plateaus, demonstrating that the model is generalizing well to unseen data without overfitting.

4.2. Direct Preference Optimization (DPO)

The second stage involved DPO training over 5 epochs using preference pairs constructed from reasoning-based (“chosen”) and generic (“rejected”) responses. The model attained a final training loss of **0.1681** and an evaluation loss of **0.0324**, with a runtime of 6 min 42.51 sec for training and 4.10 sec for evaluation, confirming the efficiency of the pipeline. This aligns with recent findings that DPO effectively transforms LLMs into implicit reward models, enabling robust preference alignment [3].

The corresponding loss trajectories are shown in Figure 5, while the training accuracies are presented in Figure 6, showing the model’s robust performance in correctly classifying the preference pairs.

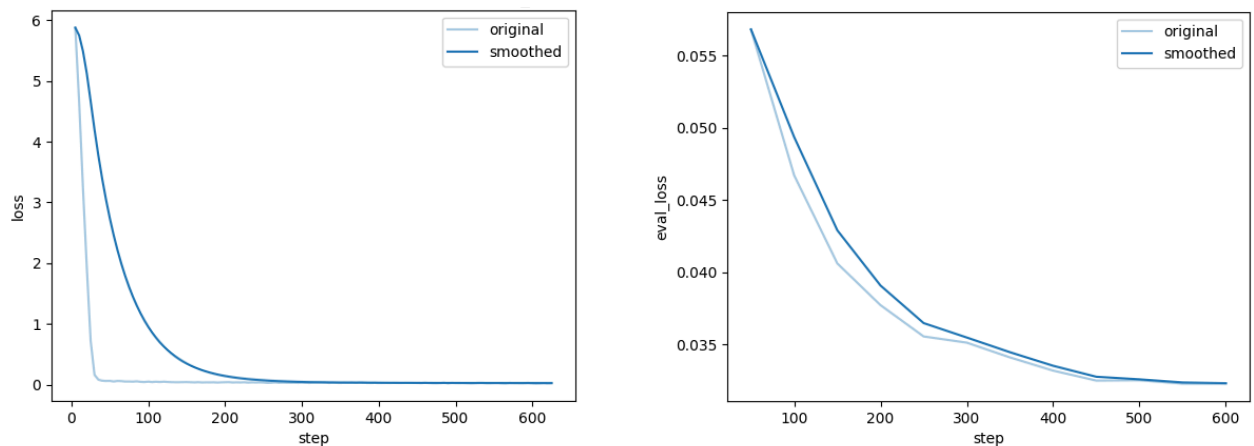


Figure 5. DPO training and evaluation loss curves. The plot on the left shows the training loss, which rapidly decreases and converges to a near-zero value. The plot on the right displays the evaluation loss, which also shows a consistent and smooth decline, indicating that the DPO fine-tuning process is stable and effectively generalizing to unseen preference pairs.

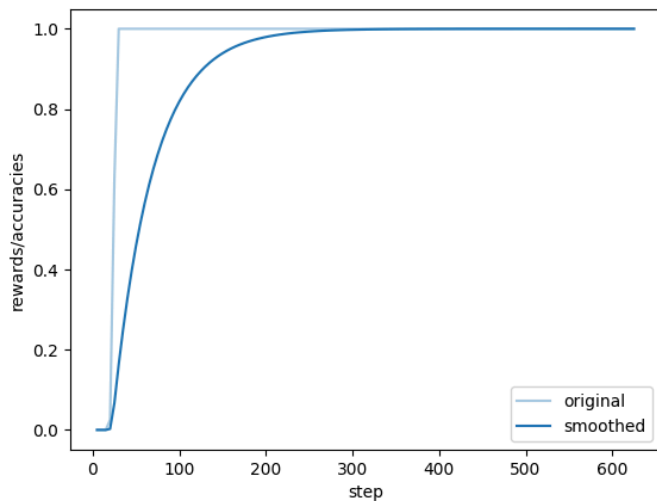


Figure 6. DPO Training and Evaluation Accuracies. The plot shows the model's accuracy in correctly classifying chosen responses over rejected ones during DPO training. The accuracy rapidly rises and converges to 100%, indicating successful alignment with the preference data.

4.3. Test Set Performance

To rigorously evaluate our method, we conducted two primary analyses: first, an internal comparison to measure the progressive impact of each training stage, and second, a direct benchmark against the state-of-the-art MedVLM-R1 model [1].

4.3.1. Progressive Improvement from SFT and DPO

We first assessed how each component of our pipeline contributed to the final performance on a test set of 150 samples per modality. The results show a clear and significant improvement at each step.

The base model, without any fine-tuning, performed at near-random chance across all modalities, consistent with prior reports on the difficulty of zero-shot reasoning in specialized domains [7]. As visualized in Figure 7, the introduction of Supervised Fine-Tuning (SFT) led to a dramatic increase in performance, particularly for the in-domain MRI dataset (98.00%). While SFT provided a strong baseline, the subsequent alignment with Direct Preference Optimization (DPO) was crucial for enhancing out-of-domain generalization, pushing accuracy on both CT and X-ray datasets to over 91%, in line with recent findings on preference-based optimization improving reasoning robustness [3].

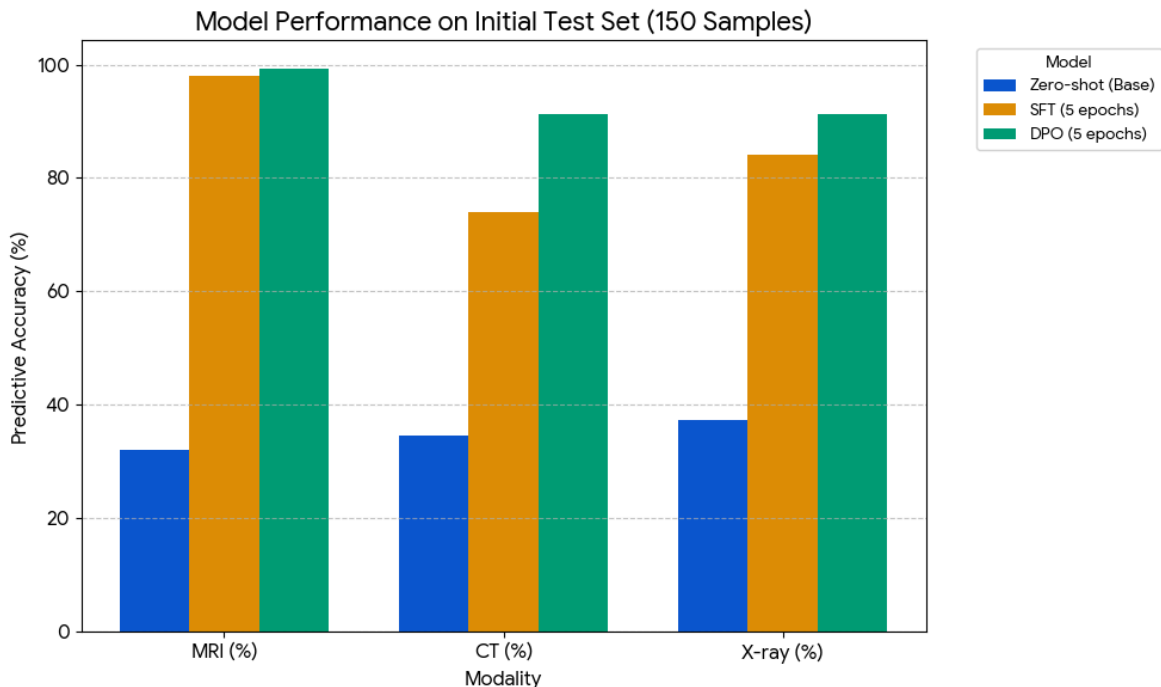


Figure 7. Accuracy comparison across training stages on in-domain (MRI) and out-of-domain (CT, X-ray) modalities. The chart highlights the substantial performance gains from Supervised Fine-Tuning (SFT) and the further generalization improvements from Direct Preference Optimization (DPO).

Model	MRI (%)	CT (%)	X-ray (%)
Zero-shot (Base)	32.00	34.67	37.33
SFT (5 epochs)	98.00	74.00	84.00
DPO (5 epochs)	99.33	91.33	91.33

Table 1. Test set performance on in-domain and out-of-domain modalities (150 samples per modality). Predictive accuracy (%) of the base model, the SFT-tuned model, and the DPO-aligned model.

4.3.2. Benchmarking Against State-of-the-Art

To contextualize our results, we benchmarked our final DPO-aligned model against MedVLM-R1 on a larger test set of 300 samples per modality. As shown in Figure 8, our approach achieves performance that is directly comparable to the MedVLM-R1 benchmark [1]. Our model matched the performance on MRI and CT and had only a marginal difference on the X-ray dataset (detailed in Table 2).

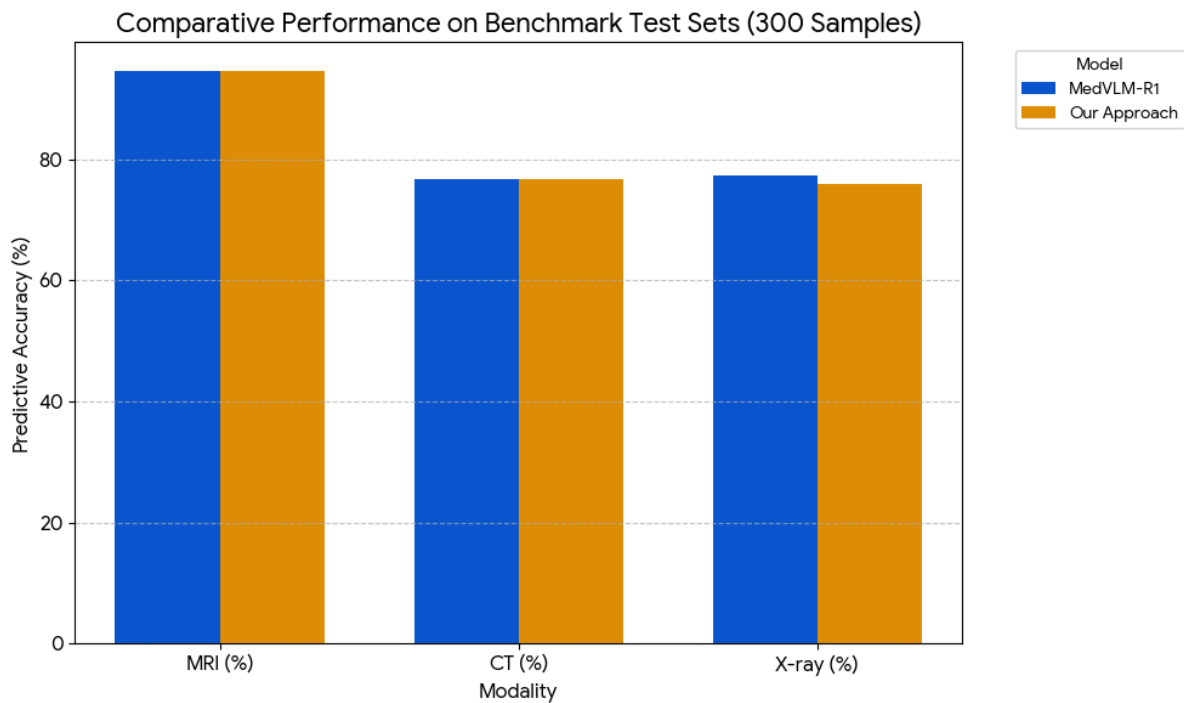


Figure 8. Comparative evaluation of our DPO-aligned approach against the MedVLM-R1 benchmark. The results show performance parity across all three modalities, validating the effectiveness of our computationally efficient method.

Model	MRI (%)	CT (%)	X-ray (%)
MedVLM-R1	94.67	76.67	77.33
Our Approach	94.67	76.67	76.00

Table 2. Comparative evaluation on benchmark test sets (300 samples per modality). Predictive accuracy (%) of MedVLM-R1 and our DPO-aligned approach.

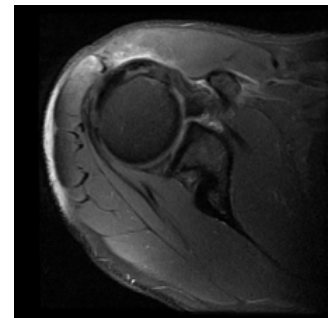
4.4. Qualitative Analysis

To highlight the reasoning capability of our aligned model, we present representative examples from each modality (MRI, CT, and X-ray). For each case, the clinical image and the ground truth are shown on the right, while the question, candidate options, and the model’s answer, are displayed on the left.

Example 1: MRI

What is the abnormality that can be seen in this image?

- A) Cartilage degeneration*
- B) Tendonitis*
- C) Nerve compression*
- D) Labral pathology*



Model Output:

<think>The MRI image shows a shoulder joint with an abnormality in the rotator cuff region. The rotator cuff is a group of muscles and tendons that help stabilize the shoulder joint and provide movement. The abnormality appears as a soft tissue mass, which is characteristic of labral pathology.**</think>**

<answer>D**</answer>**

Ground Truth: D

Example 2: CT

What is the term referring to the abnormality observed in the image?

- A) Airspace opacity*
- B) Pleural effusion*
- C) Lung consolidation*
- D) Atelectasis*



Model Output:

<think>The image shows a cross-sectional view of the chest, which is a common imaging technique used in medical diagnostics. The abnormality observed in the image appears as an area of increased density, which is characteristic of a consolidation or a mass. The presence of airspaces and the normal appearance of the pleura suggest that this is not a pleural effusion or a mass effect. The absence of signs of atelectasis (e.g., air trapping) makes it less likely to be a mass effect.</think>

<answer>A</answer>

Ground Truth: A

Example 3: X-ray

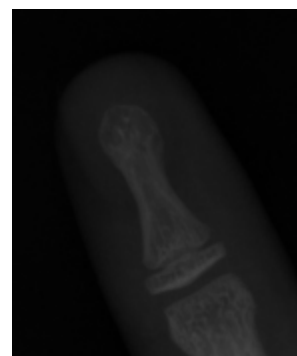
What is the name of the joint from which the bone in the picture originates?

A) Glenohumeral Connection

B) Sacroiliac Articulation

C) Radius

D) Distal Interphalangeal



Model Output:

<think>The image shows a bone with a fracture, and the bone appears to be from a human finger. The fracture is located at the joint where the bone meets the finger. To determine the name of the joint, let's analyze the image and the bones present. The bone in the image is a long bone, likely from the finger, given its shape and the context of the fracture. The fracture is at the joint where the bone meets the finger, which is the distal interphalangeal (DIP) joint.</think>

<answer>D</answer>

Ground Truth: D

4.5. Discussion

Overall, our experiments demonstrate that supervised fine-tuning provides substantial gains over the zero-shot baseline, particularly for in-domain MRI reasoning, while alignment through DPO further improves generalization to out-of-domain modalities such as CT and X-ray [3][4]. Despite training on a significantly smaller scale and with limited resources, our approach achieves performance comparable to MedVLM-R1 in both accuracy and reasoning quality [1], while reducing training time by an order of magnitude. The qualitative case studies further highlight the enhanced interpretability of DPO-aligned responses, showing more clinically grounded reasoning. These results collectively underline the promise of lightweight preference

optimization as an efficient pathway toward domain-specific alignment of multimodal LLMs [2][9][10][11].

The practical implications of this efficiency are significant. By lowering the barrier to entry for institutions with limited computational resources, our DPO-based pipeline democratizes the development of specialized medical AI [5][6]. Furthermore, the improvement in reasoning quality moves beyond mere accuracy gains. By directly optimizing for clinically sound, step-by-step explanations, this approach fosters the transparency and trustworthiness required for real-world adoption [3][12]. A model that can articulate how it arrived at a diagnosis is no longer just a predictive tool but a potential collaborative partner for clinicians, addressing a critical hurdle for deploying AI in high-stakes medical settings [8].

5. Limitations

Although the proposed pipeline demonstrates efficiency and strong reasoning performance, several limitations should be acknowledged. First, the construction of preference pairs requires access to high-quality chain-of-thought annotations from domain experts. While effective in improving reasoning, such annotations are time-consuming to obtain and may limit scalability when extending to larger or more diverse datasets [3].

Second, our evaluation was performed on relatively small curated datasets. While these provide useful insights, they do not fully capture the complexity of real-world clinical practice, where image quality, patient demographics, and disease distributions are highly variable [11].

Third, we focused primarily on accuracy and reasoning alignment; other critical aspects such as robustness to noisy inputs, fairness across subpopulations, and explainability of generated reasoning were not extensively analyzed [4][8]. Finally, translation of these findings into clinical deployment will require further validation under strict regulatory, safety, and ethical guidelines to ensure trustworthy adoption [1][9].

6. Conclusion

We presented an efficient alignment strategy for medical visual question answering by combining supervised fine-tuning with direct preference optimization [3][4]. Our approach matches MedVLM-R1 accuracy [1] while reducing training time by over an order of magnitude and requiring far fewer resources [5][6]. Beyond efficiency, DPO improves reasoning quality by learning directly from preferred chain-of-thought responses, enabling more reliable and clinically meaningful outputs [9][10][11].

This work demonstrates that lightweight preference alignment can make trustworthy medical VLMs achievable even in low-resource clinical settings [2][12]. More broadly, strategies like DPO can enhance safety, interpretability, and robustness in domains where reliability is critical.

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